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Title: Write a script in Pig to join, group, sort and filter data.

Outcome: Students will be able to write a script in Pig to join, group, sort and filter

data. **Theory:**

Apache Pig - Group Operator

The **GROUP** operator is used to group the data in one or more relations. It collects the data having the same key.

Syntax

Given below is the syntax of the **group** operator.

```
grunt> Group_data = GROUP Relation_name BY age;
```

Example

Assume that we have a file named **student_details.txt** in the HDFS directory **/pig_data/** as shown below.

student_details.txt

```
001,Rajiv,Reddy,21,9848022337,Hyderabad
002,siddarth,Battacharya,22,9848022338,Kolkata
003,Rajesh,Khanna,22,9848022339,Delhi
004,Preethi,Agarwal,21,9848022330,Pune
005,Trupthi,Mohanthy,23,9848022336,Bhuwaneshwar
006,Archana,Mishra,23,9848022335,Chennai
007,Komal,Nayak,24,9848022334,trivendram
008,Bharathi,Nambiayar,24,9848022333,Chennai
```

And we have loaded this file into Apache Pig with the relation name **student_details** as shown below.

```
grunt> student_details = LOAD 'hdfs://localhost:9000/pig_data/student_details.txt'
USING  PigStorage(',')
 as (id:int, firstname:chararray, lastname:chararray, age:int, phone:chararray, city:chararray);
```

Now, let us group the records/tuples in the relation by age as shown below.

```
grunt> group_data = GROUP student_details by age;
```

Verification

Verify the relation **group_data** using the **DUMP** operator as shown below.

```
grunt> Dump group_data;
```

Output

Then you will get output displaying the contents of the relation named **group_data** as shown below. Here you can observe that the resulting schema has two columns −

- One is **age**, by which we have grouped the relation.
- The other is a **bag**, which contains the group of tuples, student records with the respective age.

```
(21,{(4,Preethi,Agarwal,21,9848022330,Pune),(1,Rajiv,Reddy,21,9848022337,Hydera
bad)})
(22,{(3,Rajesh,Khanna,22,9848022339,Delhi),(2,siddarth,Battacharya,22,984802233
8,Kolkata)})
(23,{(6,Archana,Mishra,23,9848022335,Chennai),(5,Trupthi,Mohanthy,23,9848022336
,Bhuwaneshwar)})
(24,{(8,Bharathi,Nambiayar,24,9848022333,Chennai),(7,Komal,Nayak,24,9848022334,
trivendram)})
```

You can see the schema of the table after grouping the data using the **describe** command as shown below.

```
grunt> Describe group_data;
```

```
group_data: {group: int,student_details: {(id: int,firstname: chararray,
 lastname: chararray,age: int,phone: chararray,city: chararray)}}
```

In the same way, you can get the sample illustration of the schema using the **illustrate** command as shown below.

```
$ Illustrate group_data;
```

It will produce the following output −

```
-------------------------------------------------------------------------------------------
|group_data| group:int |
student_details:bag{:tuple(id:int,firstname:chararray,lastname:chararray,age:int,phone:charar
ray,city:chararray)}|
------------------------------------------------------------------------------------------- | | 21
| { 4, Preethi, Agarwal, 21, 9848022330, Pune), (1, Rajiv, Reddy, 21,  9848022337,
Hyderabad)}|
| | 2 |
 {(2,siddarth,Battacharya,22,9848022338,Kolkata),(003,Rajesh,Khanna,22,9848022339,Delhi
)}|
-------------------------------------------------------------------------------------------
```

Grouping by Multiple Columns

Let us group the relation by age and city as shown below.

```
grunt> group_multiple = GROUP student_details by (age, city);
```

You can verify the content of the relation named **group_multiple** using the Dump operator as shown below.

grunt> Dump group_multiple;

((21,Pune),{(4,Preethi,Agarwal,21,9848022330,Pune)})
((21,Hyderabad),{(1,Rajiv,Reddy,21,9848022337,Hyderabad)})
((22,Delhi),{(3,Rajesh,Khanna,22,9848022339,Delhi)})
((22,Kolkata),{(2,siddarth,Battacharya,22,9848022338,Kolkata)})
((23,Chennai),{(6,Archana,Mishra,23,9848022335,Chennai)})
((23,Bhuwaneshwar),{(5,Trupthi,Mohanthy,23,9848022336,Bhuwaneshwar)})
((24,Chennai),{(8,Bharathi,Nambiayar,24,9848022333,Chennai)})
(24,trivendram),{(7,Komal,Nayak,24,9848022334,trivendram)})

Group All

You can group a relation by all the columns as shown below.

grunt> **group_all** = GROUP **student_details** All;

Now, verify the content of the relation **group_all** as shown below.

grunt> Dump group_all;

(all,{(8,Bharathi,Nambiayar,24,9848022333,Chennai),(7,Komal,Nayak,24,9848022334
,trivendram),
(6,Archana,Mishra,23,9848022335,Chennai),(5,Trupthi,Mohanthy,23,9848022336,Bhuw
aneshwar),
(4,Preethi,Agarwal,21,9848022330,Pune),(3,Rajesh,Khanna,22,9848022339,Delhi),
(2,siddarth,Battacharya,22,9848022338,Kolkata),(1,Rajiv,Reddy,21,9848022337,Hyd
erabad)})

Apache Pig - Join Operator

The **JOIN** operator is used to combine records from two or more relations. While performing a join operation, we declare one (or a group of) tuple(s) from each relation, as keys. When these keys match, the two particular tuples are matched, else the records are dropped. Joins can be of the following types −

- Self-join
- Inner-join – (Equi Join) – No. of rows matching in both tables.
- Outer-join − left join, right join, and full join

This chapter explains with examples how to use the join operator in Pig Latin. Assume that we have two files namely **customers.txt** and **orders.txt** in the **/pig_data/** directory of HDFS as shown below.

customers.txt

1,Ramesh,32,Ahmedabad,2000.00
2,Khilan,25,Delhi,1500.00
3,kaushik,23,Kota,2000.00
4,Chaitali,25,Mumbai,6500.00
5,Hardik,27,Bhopal,8500.00

6,Komal,22,MP,4500.00
7,Muffy,24,Indore,10000.00

orders.txt

102,2009-10-08 00:00:00,3,3000
100,2009-10-08 00:00:00,3,1500
101,2009-11-20 00:00:00,2,1560
103,2008-05-20 00:00:00,4,2060

And we have loaded these two files into Pig with the relations **customers** and **orders** as shown below.

```
grunt> customers = LOAD 'hdfs://localhost:9000/pig_data/customers.txt'
USING  PigStorage(',')
 as (id:int, name:chararray, age:int, address:chararray, salary:int);

grunt> orders = LOAD 'hdfs://localhost:9000/pig_data/orders.txt' USING
PigStorage(',')  as (oid:int, date:chararray, customer_id:int, amount:int);
```

Let us now perform various Join operations on these two relations.

Self - join

Self-join is used to join a table with itself as if the table were two relations, temporarily renaming at least one relation.

Generally, in Apache Pig, to perform self-join, we will load the same data multiple times, under different aliases (names). Therefore let us load the contents of the file **customers.txt** as two tables as shown below.

```
grunt> customers1 = LOAD 'hdfs://localhost:9000/pig_data/customers.txt'
USING  PigStorage(',')
 as (id:int, name:chararray, age:int, address:chararray, salary:int);

grunt> customers2 = LOAD 'hdfs://localhost:9000/pig_data/customers.txt'
USING  PigStorage(',')
 as (id:int, name:chararray, age:int, address:chararray, salary:int);
```

Syntax

Given below is the syntax of performing **self-join** operation using the **JOIN** operator.

```
grunt> Relation3_name = JOIN Relation1_name BY key, Relation2_name BY key ;
```

Example

Let us perform **self-join** operation on the relation **customers**, by joining the two relations **customers1** and **customers2** as shown below.

```
grunt> customers3 = JOIN customers1 BY id, customers2 BY id;
```

Verification

Verify the relation **customers3** using the **DUMP** operator as shown below.

```
grunt> Dump customers3;
```

Output

It will produce the following output, displaying the contents of the relation **customers**.

```
(1,Ramesh,32,Ahmedabad,2000,1,Ramesh,32,Ahmedabad,2000)
(2,Khilan,25,Delhi,1500,2,Khilan,25,Delhi,1500)
(3,kaushik,23,Kota,2000,3,kaushik,23,Kota,2000)
(4,Chaitali,25,Mumbai,6500,4,Chaitali,25,Mumbai,6500)
(5,Hardik,27,Bhopal,8500,5,Hardik,27,Bhopal,8500)
(6,Komal,22,MP,4500,6,Komal,22,MP,4500)
(7,Muffy,24,Indore,10000,7,Muffy,24,Indore,10000)
```

Inner Join

Inner Join is used quite frequently; it is also referred to as **equijoin**. An inner join returns rows when there is a match in both tables.

It creates a new relation by combining column values of two relations (say A and B) based upon the join-predicate. The query compares each row of A with each row of B to find all pairs of rows which satisfy the join-predicate. When the join-predicate is satisfied, the column values for each matched pair of rows of A and B are combined into a result row.

Syntax

Here is the syntax of performing **inner join** operation using the **JOIN** operator.

```
grunt> result = JOIN relation1 BY columnname, relation2 BY columnname;
```

Example

Let us perform **inner join** operation on the two relations **customers** and **orders** as shown below.

```
grunt> coustomer_orders = JOIN customers BY id, orders BY customer_id;
```

Verification

Verify the relation **coustomer_orders** using the **DUMP** operator as shown below.

```
grunt> Dump coustomer_orders;
```

Output

You will get the following output that will the contents of the relation named **coustomer_orders**.

```
(2,Khilan,25,Delhi,1500,101,2009-11-20 00:00:00,2,1560)
(3,kaushik,23,Kota,2000,100,2009-10-08 00:00:00,3,1500)
(3,kaushik,23,Kota,2000,102,2009-10-08 00:00:00,3,3000)
(4,Chaitali,25,Mumbai,6500,103,2008-05-20 00:00:00,4,2060)
```

Note −

***Outer Join*:** Unlike inner join, **outer join** returns all the rows from at least one of the relations. An outer join operation is carried out in three ways −

- Left outer join
- Right outer join
- Full outer join

Left Outer Join

The **left outer Join** operation returns all rows from the left table, even if there are no matches in the right relation.

Syntax

Given below is the syntax of performing **left outer join** operation using the **JOIN** operator.

```
grunt> Relation3_name = JOIN Relation1_name BY id LEFT OUTER, Relation2_name BY customer_id;
```

Example

Let us perform left outer join operation on the two relations customers and orders as shown below.

```
grunt> outer_left = JOIN customers BY id LEFT OUTER, orders BY customer_id;
```

Verification

Verify the relation **outer_left** using the **DUMP** operator as shown below.

```
grunt> Dump outer_left;
```

Output

It will produce the following output, displaying the contents of the relation **outer_left**.

```
(1,Ramesh,32,Ahmedabad,2000,,,,)
(2,Khilan,25,Delhi,1500,101,2009-11-20 00:00:00,2,1560)
(3,kaushik,23,Kota,2000,100,2009-10-08 00:00:00,3,1500)
(3,kaushik,23,Kota,2000,102,2009-10-08 00:00:00,3,3000)
(4,Chaitali,25,Mumbai,6500,103,2008-05-20 00:00:00,4,2060)
(5,Hardik,27,Bhopal,8500,,,,)
(6,Komal,22,MP,4500,,,,)
(7,Muffy,24,Indore,10000,,,,)
```

Right Outer Join

The **right outer join** operation returns all rows from the right table, even if there are no matches in the left table.

Syntax

Given below is the syntax of performing **right outer join** operation using the **JOIN** operator.

```
grunt> outer_right = JOIN customers BY id RIGHT, orders BY customer_id;
```

Example

Let us perform **right outer join** operation on the two relations **customers** and **orders** as shown below.

```
grunt> outer_right = JOIN customers BY id RIGHT, orders BY customer_id;
```

Verification

Verify the relation **outer_right** using the **DUMP** operator as shown below.

```
grunt> Dump outer_right
```

Output

It will produce the following output, displaying the contents of the relation **outer_right**.

```
(2,Khilan,25,Delhi,1500,101,2009-11-20 00:00:00,2,1560)
(3,kaushik,23,Kota,2000,100,2009-10-08 00:00:00,3,1500)
(3,kaushik,23,Kota,2000,102,2009-10-08 00:00:00,3,3000)
(4,Chaitali,25,Mumbai,6500,103,2008-05-20 00:00:00,4,2060)
```

Full Outer Join

The **full outer join** operation returns rows when there is a match in one of the relations.

Syntax

Given below is the syntax of performing **full outer join** using the **JOIN** operator.

```
grunt> outer_full = JOIN customers BY id FULL OUTER, orders BY customer_id;
```

Example

Let us perform **full outer join** operation on the two relations **customers** and **orders** as shown below.

```
grunt> outer_full = JOIN customers BY id FULL OUTER, orders BY customer_id;
```

Verification

Verify the relation **outer_full** using the **DUMP** operator as shown below.

```
grun> Dump outer_full;
```

Output

It will produce the following output, displaying the contents of the relation **outer_full**.

```
(1,Ramesh,32,Ahmedabad,2000,,,,)
(2,Khilan,25,Delhi,1500,101,2009-11-20 00:00:00,2,1560)
(3,kaushik,23,Kota,2000,100,2009-10-08 00:00:00,3,1500)
```

(3,kaushik,23,Kota,2000,102,2009-10-08 00:00:00,3,3000)
(4,Chaitali,25,Mumbai,6500,103,2008-05-20 00:00:00,4,2060)
(5,Hardik,27,Bhopal,8500,,,,)
(6,Komal,22,MP,4500,,,,)
(7,Muffy,24,Indore,10000,,,,)

Apache Pig - Filter Operator

The **FILTER** operator is used to select the required tuples from a relation based on a condition. Syntax

Given below is the syntax of the **FILTER** operator.

```
grunt> Relation2_name = FILTER Relation1_name BY (condition);
```

Example

Assume that we have a file named **student_details.txt** in the HDFS directory **/pig_data/** as shown below.

student_details.txt

001,Rajiv,Reddy,21,9848022337,Hyderabad
002,siddarth,Battacharya,22,9848022338,Kolkata
003,Rajesh,Khanna,22,9848022339,Delhi
004,Preethi,Agarwal,21,9848022330,Pune
005,Trupthi,Mohanthy,23,9848022336,Bhuwaneshwar
006,Archana,Mishra,23,9848022335,Chennai
007,Komal,Nayak,24,9848022334,trivendram
008,Bharathi,Nambiayar,24,9848022333,Chennai

And we have loaded this file into Pig with the relation name **student_details** as shown below.

```
grunt> student_details = LOAD 'hdfs://localhost:9000/pig_data/student_details.txt'
USING PigStorage(',')
 as (id:int, firstname:chararray, lastname:chararray, age:int, phone:chararray, city:chararray);
```

Let us now use the Filter operator to get the details of the students who belong to the city Chennai.

```
filter_data = FILTER student_details BY city == 'Chennai';
```

Verification

Verify the relation **filter_data** using the **DUMP** operator as shown below.

```
grunt> Dump filter_data;
```

Output

It will produce the following output, displaying the contents of the relation **filter_data** as follows.

(6,Archana,Mishra,23,9848022335,Chennai)
(8,Bharathi,Nambiayar,24,9848022333,Chennai)

Results :

```
EXPERIMENT 8 : Pig script to join, group, sort and filter data
Part 1 : Input datasets and upload to HDFS

$ head -n 20 data/student_details.txt data/customers.txt data/orders.txt
==> data/student_details.txt <==
001,Rajiv,Reddy,21,9848022337,Hyderabad
002,siddarth,Battacharya,22,9848022338,Kolkata
003,Rajesh,Khanna,22,9848022339,Delhi
004,Preethi,Agarwal,21,9848022330,Pune
005,Trupthi,Mohanthy,23,9848022336,Bhuwaneshwar
006,Archana,Mishra,23,9848022335,Chennai
007,Komal,Nayak,24,9848022334,trivendram
008,Bharathi,Nambiayar,24,9848022333,Chennai

==> data/customers.txt <==
1,Ramesh,32,Ahmedabad,2000.00
2,Khilan,25,Delhi,1500.00
3,kaushik,23,Kota,2000.00
4,Chaitali,25,Mumbai,6500.00
5,Hardik,27,Bhopal,8500.00
6,Komal,22,MP,4500.00
7,Muffy,24,Indore,10000.00

==> data/orders.txt <==
102,2009-10-08 00:00:00,3,3000
100,2009-10-08 00:00:00,3,1500
101,2009-11-20 00:00:00,2,1560
103,2008-05-20 00:00:00,4,2060
$ head -n 20 data/students.txt data/departments.txt data/employees.txt
==> data/students.txt <==
1,Aarav,D01
2,Diya,D02
3,Kabir,D01
4,Meera,D03
5,Rohan,D02
6,Sneha,D04

==> data/departments.txt <==
D01,Computer Engineering
D02,Information Technology
D03,Data Science

==> data/employees.txt <==
E01,IT,65000
E02,HR,48000
E03,Finance,72000
E04,IT,81000
E05,Sales,39000
E06,HR,52000
E07,Finance,58000
E08,IT,47000
$ hdfs dfs -mkdir -p /bdt/exp8
$ hdfs dfs -put -f data/customers.txt data/departments.txt data/employees.txt data/orders.txt data/student_details.txt data/students.txt /bdt/exp8/
$ hdfs dfs -ls /bdt/exp8
Found 6 items
-rw-r--r--   1 ubuntu supergroup        187 2026-10-25 19:47 /bdt/exp8/customers.txt
-rw-r--r--   1 ubuntu supergroup         69 2026-10-25 19:47 /bdt/exp8/departments.txt
-rw-r--r--   1 ubuntu supergroup        117 2026-10-25 19:47 /bdt/exp8/employees.txt
-rw-r--r--   1 ubuntu supergroup        124 2026-10-25 19:47 /bdt/exp8/orders.txt
-rw-r--r--   1 ubuntu supergroup        339 2026-10-25 19:47 /bdt/exp8/student_details.txt
-rw-r--r--   1 ubuntu supergroup         71 2026-10-25 19:47 /bdt/exp8/students.txt
```

```
EXPERIMENT 8 : Pig script to join, group, sort and filter data
Part 2 : GROUP: by one column, by multiple columns, with COUNT, and GROUP ALL

$ cat pig/01_group.pig
-- Experiment 8 | Part 2 : GROUP operator
student_details = LOAD '/bdt/exp8/student_details.txt' USING PigStorage(',')
    AS (id:int, firstname:chararray, lastname:chararray, age:int, phone:chararray, city:chararray);

-- group by one column
group_data = GROUP student_details BY age;
DESCRIBE group_data;
DUMP group_data;

-- group by multiple columns
group_multiple = GROUP student_details BY (age, city);
DUMP group_multiple;

-- aggregate on the groups: number of students of each age
age_count = FOREACH group_data GENERATE group AS age, COUNT(student_details) AS students;
DUMP age_count;

-- group all records into a single group
group_all = GROUP student_details ALL;
DUMP group_all;

$ pig -x mapreduce -4 /home/ubuntu/bdt/pig/log4j.properties -f pig/01_group.pig
2026-10-25 19:48:26,397 INFO pig.ExecTypeProvider: Trying ExecType : LOCAL
2026-10-25 19:48:26,397 INFO pig.ExecTypeProvider: Trying ExecType : MAPREDUCE
2026-10-25 19:48:26,398 INFO pig.ExecTypeProvider: Picked MAPREDUCE as the ExecType
2026-10-25 19:48:26,427 INFO pig.Main: Loaded log4j properties from file: /home/ubuntu/bdt/pig/log4j.properties
19:48:26 [INFO] Main - Apache Pig version 0.18.0 (r1926484) compiled Jun 16 2025, 10:38:21
19:48:26 [INFO] Main - Logging error messages to: /home/ubuntu/bdt/exp8/pig_1792937906425.log
19:48:26 [INFO] HExecutionEngine - Connecting to hadoop file system at: hdfs://localhost:9000
19:48:26 [WARN] PigServer - ATS is disabled since yarn.timeline-service.enabled set to false
group_data: {group: int,student_details: {{id: int,firstname: chararray,lastname: chararray,age: int,phone: chararray,city: chararray}}}
19:48:28 [INFO] MapReduceLauncher - 1 map-reduce job(s) waiting for submission.
19:48:28 [WARN] JobResourceUploader - No job jar file set.  User classes may not be found. See Job or Job#setJar(String).
19:48:29 [INFO] MapReduceLauncher - HadoopJobId: job_1792936583670_0037
19:48:29 [INFO] MapReduceLauncher - Processing aliases group_data,student_details
19:48:29 [INFO] MapReduceLauncher - detailed locations: M: student_details[2,18],student_details[-1,-1],group_data[6,13] C:  R:
19:48:29 [INFO] MapReduceLauncher - 0% complete
19:48:29 [INFO] MapReduceLauncher - Running jobs are [job_1792936583670_0037]
19:48:37 [INFO] MapReduceLauncher - 50% complete
19:48:37 [INFO] MapReduceLauncher - Running jobs are [job_1792936583670_0037]
19:48:40 [INFO] MapReduceLauncher - Running jobs are [job_1792936583670_0037]
19:48:45 [INFO] MapReduceLauncher - 100% complete
19:48:45 [INFO] SimplePigStats - Script Statistics:

HadoopVersion   PigVersion      UserId  StartedAt       FinishedAt      Features
3.3.6   0.18.0  ubuntu  2026-10-25 19:48:27     2026-10-25 19:48:45     GROUP_BY

Success!

Job Stats (time in seconds):
JobId   Maps    Reduces MaxMapTime      MinMapTime      AvgMapTime      MedianMapTime   MaxReduceTime   MinReduceTime   AvgReduceTime   MedianReducetime        Alias   Feature Outputs
job_1792936583670_0037  1       1       1       1       1       1       1       1       1       group_data,student_details      GROUP_BY        hdfs://localhost:9000/tmp/temp-16903809/tmp654828190,

Input(s):
Successfully read 8 records (716 bytes) from: "/bdt/exp8/student_details.txt"

Output(s):
Successfully stored 4 records (416 bytes) in: "hdfs://localhost:9000/tmp/temp-16903809/tmp654828190"

Counters:
Total records written : 4
Total bytes written : 416
Spillable Memory Manager spill count : 0
Total bags proactively spilled: 0
Total records proactively spilled: 0

Job DAG:
job_1792936583670_0037


19:48:45 [INFO] MapReduceLauncher - Success!
```

```
Input(s):
Successfully read 8 records (716 bytes) from: "/bdt/exp8/student_details.txt"

Output(s):
Successfully stored 8 records (529 bytes) in: "hdfs://localhost:9000/tmp/temp-16903809/tmp-1006285349"

Counters:
Total records written : 8
Total bytes written : 529
Spillable Memory Manager spill count : 0
Total bags proactively spilled: 0
Total records proactively spilled: 0

Job DAG:
job_1792936583670_0038


19:49:03 [INFO] MapReduceLauncher - Success!
((21,Pune),{(4,Preethi,Agarwal,21,9848022330,Pune)})
((21,Hyderabad),{(1,Rajiv,Reddy,21,9848022337,Hyderabad)})
((22,Delhi),{(3,Rajesh,Khanna,22,9848022339,Delhi)})
((22,Kolkata),{(2,siddarth,Battacharya,22,9848022338,Kolkata)})
((23,Chennai),{(6,Archana,Mishra,23,9848022335,Chennai)})
((23,Bhuwaneshwar),{(5,Trupthi,Mohanthy,23,9848022336,Bhuwaneshwar)})
((24,Chennai),{(8,Bharathi,Nambiayar,24,9848022333,Chennai)})
((24,trivendram),{(7,Komal,Nayak,24,9848022334,trivendram)})
19:49:04 [INFO] MapReduceLauncher - 1 map-reduce job(s) waiting for submission.
19:49:04 [WARN] JobResourceUploader - No job jar file set.  User classes may not be found. See Job or Job#setJar(String).
19:49:06 [INFO] MapReduceLauncher - HadoopJobId: job_1792936583670_0039
19:49:06 [INFO] MapReduceLauncher - Processing aliases age_count,group_data,student_details
19:49:06 [INFO] MapReduceLauncher - detailed locations: M: student_details[2,18],student_details[-1,-1],age_count[15,12],group_data[6,13] C: age_count[15,12],group_data[6,13] R: age_count[15,12]
19:49:06 [INFO] MapReduceLauncher - 0% complete
19:49:06 [INFO] MapReduceLauncher - Running jobs are [job_1792936583670_0039]
19:49:13 [INFO] MapReduceLauncher - 50% complete
19:49:13 [INFO] MapReduceLauncher - Running jobs are [job_1792936583670_0039]
19:49:16 [INFO] MapReduceLauncher - Running jobs are [job_1792936583670_0039]
19:49:21 [INFO] MapReduceLauncher - 100% complete
19:49:21 [INFO] SimplePigStats - Script Statistics:

HadoopVersion   PigVersion      UserId  StartedAt       FinishedAt      Features
3.3.6   0.18.0  ubuntu  2026-10-25 19:49:03     2026-10-25 19:49:21     GROUP_BY

Success!

Job Stats (time in seconds):
JobId   Maps    Reduces MaxMapTime      MinMapTime      AvgMapTime      MedianMapTime   MaxReduceTime   MinReduceTime   AvgReduceTime   MedianReducetime        Alias   Feature Outputs
job_1792936583670_0039  1       1       1       1       1       1       1       1       age_count,group_data,student_details    GROUP_BY,COMBINER       hdfs://localhost:9000/tmp/temp-16903809/tmp-349451843,

Input(s):
Successfully read 8 records (716 bytes) from: "/bdt/exp8/student_details.txt"

Output(s):
Successfully stored 4 records (30 bytes) in: "hdfs://localhost:9000/tmp/temp-16903809/tmp-349451843"

Counters:
Total records written : 4
Total bytes written : 30
Spillable Memory Manager spill count : 0
Total bags proactively spilled: 0
Total records proactively spilled: 0

Job DAG:
job_1792936583670_0039


19:49:21 [INFO] MapReduceLauncher - Success!
(21,2)
(22,2)
(23,2)
(24,2)
19:49:22 [INFO] MapReduceLauncher - 1 map-reduce job(s) waiting for submission.
19:49:22 [WARN] JobResourceUploader - No job jar file set.  User classes may not be found. See Job or Job#setJar(String).
19:49:23 [INFO] MapReduceLauncher - HadoopJobId: job_1792936583670_0040
19:49:23 [INFO] MapReduceLauncher - Processing aliases group_all,student_details
19:49:23 [INFO] MapReduceLauncher - detailed locations: M: student_details[2,18],student_details[-1,-1],group_all[19,12] C:  R:
```

```
3.3.6   0.18.0  ubuntu  2026-10-25 19:49:03     2026-10-25 19:49:21     GROUP_BY

Success!

Job Stats (time in seconds):
JobId   Maps    Reduces MaxMapTime      MinMapTime      AvgMapTime      MedianMapTime   MaxReduceTime   MinReduceTime   AvgReduceTime   MedianReducetime        Alias   Feature Outputs
job_1792936583670_0039  1       1       1       1       1       1       1       1       age_count,group_data,student_details    GROUP_BY,COMBINER       hdfs://localhost:9000/tmp/temp-16903809/tmp-349451843,

Input(s):
Successfully read 8 records (716 bytes) from: "/bdt/exp8/student_details.txt"

Output(s):
Successfully stored 4 records (30 bytes) in: "hdfs://localhost:9000/tmp/temp-16903809/tmp-349451843"

Counters:
Total records written : 4
Total bytes written : 30
Spillable Memory Manager spill count : 0
Total bags proactively spilled: 0
Total records proactively spilled: 0

Job DAG:
job_1792936583670_0039


19:49:21 [INFO] MapReduceLauncher - Success!
(21,2)
(22,2)
(23,2)
(24,2)
19:49:22 [INFO] MapReduceLauncher - 1 map-reduce job(s) waiting for submission.
19:49:22 [WARN] JobResourceUploader - No job jar file set.  User classes may not be found. See Job or Job#setJar(String).
19:49:23 [INFO] MapReduceLauncher - HadoopJobId: job_1792936583670_0040
19:49:23 [INFO] MapReduceLauncher - Processing aliases group_all,student_details
19:49:23 [INFO] MapReduceLauncher - detailed locations: M: student_details[2,18],student_details[-1,-1],group_all[19,12] C:  R:
19:49:23 [INFO] MapReduceLauncher - 0% complete
19:49:23 [INFO] MapReduceLauncher - Running jobs are [job_1792936583670_0040]
19:49:30 [INFO] MapReduceLauncher - 50% complete
19:49:30 [INFO] MapReduceLauncher - Running jobs are [job_1792936583670_0040]
19:49:35 [INFO] MapReduceLauncher - Running jobs are [job_1792936583670_0040]
19:49:38 [INFO] MapReduceLauncher - 100% complete
19:49:38 [INFO] SimplePigStats - Script Statistics:

HadoopVersion   PigVersion      UserId  StartedAt       FinishedAt      Features
3.3.6   0.18.0  ubuntu  2026-10-25 19:49:21     2026-10-25 19:49:38     GROUP_BY

Success!

Job Stats (time in seconds):
JobId   Maps    Reduces MaxMapTime      MinMapTime      AvgMapTime      MedianMapTime   MaxReduceTime   MinReduceTime   AvgReduceTime   MedianReducetime        Alias   Feature Outputs
job_1792936583670_0040  1       1       1       1       1       1       1       1       group_all,student_details       GROUP_BY        hdfs://localhost:9000/tmp/temp-16903809/tmp-1767758806,

Input(s):
Successfully read 8 records (716 bytes) from: "/bdt/exp8/student_details.txt"

Output(s):
Successfully stored 1 records (405 bytes) in: "hdfs://localhost:9000/tmp/temp-16903809/tmp-1767758806"

Counters:
Total records written : 1
Total bytes written : 405
Spillable Memory Manager spill count : 0
Total bags proactively spilled: 0
Total records proactively spilled: 0

Job DAG:
job_1792936583670_0040


19:49:38 [INFO] MapReduceLauncher - Success!
(all,{(8,Bharathi,Nambiayar,24,9848022333,Chennai),(7,Komal,Nayak,24,9848022334,trivendram),(6,Archana,Mishra,23,9848022335,Chennai),(5,Trupthi,Mohanthy,23,9848022336,Bhuwaneshwar),(4,Preethi,Agarwal,21,9848022330,Pune),(3,Rajesh,Khanna,22,9
848022339,Delhi),(2,siddarth,Battacharya,22,9848022338,Kolkata),(1,Rajiv,Reddy,21,9848022337,Hyderabad)})
19:49:38 [INFO] Main - Pig script completed in 1 minute, 12 seconds and 450 milliseconds (72450 ms)

{ screenshot 2 - take it, then press Enter }
```

```
EXPERIMENT 8 : Pig script to join, group, sort and filter data
Part 3 : JOIN: self join and inner (equi) join

$ cat pig/02_join_self_inner.pig
-- Experiment 8 | Part 3 : JOIN - self join and inner (equi) join
customers1 = LOAD '/bdt/exp8/customers.txt' USING PigStorage(',')
   AS (id:int, name:chararray, age:int, address:chararray, salary:double);
customers2 = LOAD '/bdt/exp8/customers.txt' USING PigStorage(',')
   AS (id:int, name:chararray, age:int, address:chararray, salary:double);
orders     = LOAD '/bdt/exp8/orders.txt' USING PigStorage(',')
   AS (oid:int, date:chararray, customer_id:int, amount:int);

-- self join: the same data loaded under two aliases
customers3 = JOIN customers1 BY id, customers2 BY id;
DUMP customers3;

-- inner join: only customers that have orders
customer_orders = JOIN customers1 BY id, orders BY customer_id;
DESCRIBE customer_orders;
DUMP customer_orders;

$ pig -x mapreduce -4 /home/ubuntu/bdt/pig/log4j.properties -f pig/02_join_self_inner.pig
2026-10-25 19:50:02,817 INFO pig.ExecTypeProvider: Trying ExecType : LOCAL
2026-10-25 19:50:02,818 INFO pig.ExecTypeProvider: Trying ExecType : MAPREDUCE
2026-10-25 19:50:02,818 INFO pig.ExecTypeProvider: Picked MAPREDUCE as the ExecType
2026-10-25 19:50:02,844 INFO pig.Main: Loaded log4j properties from file: /home/ubuntu/bdt/pig/log4j.properties
19:50:02 [INFO] Main - Apache Pig version 0.18.0 (r1926484) compiled Jun 16 2025, 10:38:21
19:50:02 [INFO] Main - Logging error messages to: /home/ubuntu/bdt/exp8/pig_1792936002842.log
19:50:02 [INFO] HExecutionEngine - Connecting to hadoop file system at: hdfs://localhost:9000
19:50:03 [WARN] PigServer - ATS is disabled since yarn.timeline-service.enabled set to false
19:50:05 [INFO] MapReduceLauncher - 1 map-reduce job(s) waiting for submission.
19:50:05 [WARN] JobResourceUploader - No job jar file set.  User classes may not be found. See Job or Job#setJar(String).
19:50:06 [INFO] MapReduceLauncher - HadoopJobId: job_1792936583670_0041
19:50:06 [INFO] MapReduceLauncher - Processing aliases customers1,customers2,customers3
19:50:06 [INFO] MapReduceLauncher - detailed locations: M: customers1[2,13],customers1[-1,-1],customers3[10,13],customers2[4,13],customers2[-1,-1],customers3[10,13] C:  R:
19:50:06 [INFO] MapReduceLauncher - 0% complete
19:50:06 [INFO] MapReduceLauncher - Running jobs are [job_1792936583670_0041]
19:50:15 [INFO] MapReduceLauncher - 50% complete
19:50:15 [INFO] MapReduceLauncher - Running jobs are [job_1792936583670_0041]
19:50:18 [INFO] MapReduceLauncher - Running jobs are [job_1792936583670_0041]
19:50:21 [INFO] MapReduceLauncher - 100% complete
19:50:21 [INFO] SimplePigStats - Script Statistics:

HadoopVersion   PigVersion      UserId  StartedAt       FinishedAt      Features
3.3.6   0.18.0  ubuntu  2026-10-25 19:50:03     2026-10-25 19:50:21     HASH_JOIN

Success!

Job Stats (time in seconds):
JobId   Maps    Reduces MaxMapTime      MinMapTime      AvgMapTime      MedianMapTime   MaxReduceTime   MinReduceTime   AvgReduceTime   MedianReducetime        Alias   Feature Outputs
job_1792936583670_0041  2       1       2       2       2       1       1       1       customers1,customers2,customers3        HASH_JOIN       hdfs://localhost:9000/tmp/temp-555809009/tmp1079756926,

Input(s):
Successfully read 7 records from: "/bdt/exp8/customers.txt"
Successfully read 7 records from: "/bdt/exp8/customers.txt"

Output(s):
Successfully stored 7 records (450 bytes) in: "hdfs://localhost:9000/tmp/temp-555809009/tmp1079756926"

Counters:
Total records written : 7
Total bytes written : 450
Spillable Memory Manager spill count : 0
Total bags proactively spilled: 0
Total records proactively spilled: 0

Job DAG:
job_1792936583670_0041


19:50:21 [INFO] MapReduceLauncher - Success!
(1,Ramesh,32,Ahmedabad,2000.0,1,Ramesh,32,Ahmedabad,2000.0)
(2,Khilan,25,Delhi,1500.0,2,Khilan,25,Delhi,1500.0)
(3,kaushik,23,Kota,2000.0,3,kaushik,23,Kota,2000.0)
(4,Chaitali,25,Mumbai,6500.0,4,Chaitali,25,Mumbai,6500.0)
```

```
Successfully read 7 records from: "/bdt/exp8/customers.txt"
Successfully read 7 records from: "/bdt/exp8/customers.txt"

Output(s):
Successfully stored 7 records (450 bytes) in: "hdfs://localhost:9000/tmp/temp-555809009/tmp1079756926"

Counters:
Total records written : 7
Total bytes written : 450
Spillable Memory Manager spill count : 0
Total bags proactively spilled: 0
Total records proactively spilled: 0

Job DAG:
job_1792936583670_0041


19:50:21 [INFO] MapReduceLauncher - Success!
(1,Ramesh,32,Ahmedabad,2000.0,1,Ramesh,32,Ahmedabad,2000.0)
(2,Khilan,25,Delhi,1500.0,2,Khilan,25,Delhi,1500.0)
(3,kaushik,23,Kota,2000.0,3,kaushik,23,Kota,2000.0)
(4,Chaitali,25,Mumbai,6500.0,4,Chaitali,25,Mumbai,6500.0)
(5,Hardik,27,Bhopal,8500.0,5,Hardik,27,Bhopal,8500.0)
(6,Komal,22,MP,4500.0,6,Komal,22,MP,4500.0)
(7,Muffy,24,Indore,10000.0,7,Muffy,24,Indore,10000.0)
customer_orders: {customers1::id: int,customers1::name: chararray,customers1::age: int,customers1::address: chararray,customers1::salary: double,orders::oid: int,orders::date: chararray,orders::customer_id: int,orders::amount: int}
19:50:21 [WARN] SchemaTupleBackend - SchemaTupleBackend has already been initialized
19:50:22 [INFO] MapReduceLauncher - 1 map-reduce job(s) waiting for submission.
19:50:22 [WARN] JobResourceUploader - No job jar file set.  User classes may not be found. See Job or Job#setJar(String).
19:50:23 [INFO] MapReduceLauncher - HadoopJobId: job_1792936583670_0042
19:50:23 [INFO] MapReduceLauncher - Processing aliases customer_orders,customers1,orders
19:50:23 [INFO] MapReduceLauncher - detailed locations: M: customers1[2,13],customers1[-1,-1],customer_orders[14,18],orders[6,13],orders[-1,-1],customer_orders[14,18] C:  R:
19:50:23 [INFO] MapReduceLauncher - 0% complete
19:50:23 [INFO] MapReduceLauncher - Running jobs are [job_1792936583670_0042]
19:50:32 [INFO] MapReduceLauncher - 50% complete
19:50:32 [INFO] MapReduceLauncher - Running jobs are [job_1792936583670_0042]
19:50:38 [INFO] MapReduceLauncher - Running jobs are [job_1792936583670_0042]
19:50:44 [INFO] MapReduceLauncher - 100% complete
19:50:44 [INFO] SimplePigStats - Script Statistics:

HadoopVersion   PigVersion      UserId  StartedAt       FinishedAt      Features
3.3.6   0.18.0  ubuntu  2026-10-25 19:50:21     2026-10-25 19:50:44     HASH_JOIN

Success!

Job Stats (time in seconds):
JobId   Maps    Reduces MaxMapTime      MinMapTime      AvgMapTime      MedianMapTime   MaxReduceTime   MinReduceTime   AvgReduceTime   MedianReducetime        Alias   Feature Outputs
job_1792936583670_0042  2       1       2       2       2       1       1       1       customer_orders,customers1,orders       HASH_JOIN       hdfs://localhost:9000/tmp/temp-555809009/tmp-235013799,

Input(s):
Successfully read 7 records from: "/bdt/exp8/customers.txt"
Successfully read 4 records from: "/bdt/exp8/orders.txt"

Output(s):
Successfully stored 4 records (253 bytes) in: "hdfs://localhost:9000/tmp/temp-555809009/tmp-235013799"

Counters:
Total records written : 4
Total bytes written : 253
Spillable Memory Manager spill count : 0
Total bags proactively spilled: 0
Total records proactively spilled: 0

Job DAG:
job_1792936583670_0042


19:50:44 [INFO] MapReduceLauncher - Success!
(2,Khilan,25,Delhi,1500.0,101,2009-11-20 00:00:00,2,1560)
(3,kaushik,23,Kota,2000.0,100,2009-10-08 00:00:00,3,1500)
(3,kaushik,23,Kota,2000.0,102,2009-10-08 00:00:00,3,3000)
(4,Chaitali,25,Mumbai,6500.0,103,2008-05-20 00:00:00,4,2060)
19:50:44 [INFO] Main - Pig script completed in 41 seconds and 658 milliseconds (41658 ms)

[ screenshot 3 - take it, then press Enter ]
```

```
EXPERIMENT 8 : Pig script to join, group, sort and filter data
Part 4 : JOIN: left, right and full outer joins

$ cat pig/03_outer_joins.pig
-- Experiment 8 | Part 4 : JOIN - left, right and full outer joins
customers = LOAD '/bdt/exp8/customers.txt' USING PigStorage(',')
    AS (id:int, name:chararray, age:int, address:chararray, salary:double);
orders    = LOAD '/bdt/exp8/orders.txt' USING PigStorage(',')
    AS (oid:int, date:chararray, customer_id:int, amount:int);

outer_left  = JOIN customers BY id LEFT OUTER,  orders BY customer_id;
DUMP outer_left;

outer_right = JOIN customers BY id RIGHT OUTER, orders BY customer_id;
DUMP outer_right;

outer_full  = JOIN customers BY id FULL OUTER,  orders BY customer_id;
DUMP outer_full;

$ pig -x mapreduce -4 /home/ubuntu/bdt/pig/log4j.properties -f pig/03_outer_joins.pig
2026-10-25 19:51:00,149 INFO pig.ExecTypeProvider: Trying ExecType : LOCAL
2026-10-25 19:51:00,150 INFO pig.ExecTypeProvider: Trying ExecType : MAPREDUCE
2026-10-25 19:51:00,150 INFO pig.ExecTypeProvider: Picked MAPREDUCE as the ExecType
2026-10-25 19:51:00,180 INFO pig.Main: Loaded log4j properties from file: /home/ubuntu/bdt/pig/log4j.properties
19:51:00 [INFO] Main - Apache Pig version 0.18.0 (r1926484) compiled Jun 16 2025, 10:38:21
19:51:00 [INFO] Main - Logging error messages to: /home/ubuntu/bdt/exp8/pig_1729230860178.log
19:51:00 [INFO] HExecutionEngine - Connecting to hadoop file system at: hdfs://localhost:9000
19:51:00 [WARN] PigServer - ATS is disabled since yarn.timeline-service.enabled set to false
19:51:02 [INFO] MapReduceLauncher - 1 map-reduce job(s) waiting for submission.
19:51:02 [WARN] JobResourceUploader - No job jar file set.  User classes may not be found. See Job or Job#setJar(String).
19:51:03 [INFO] MapReduceLauncher - HadoopJobId: job_1729236583670_0043
19:51:03 [INFO] MapReduceLauncher - Processing aliases customers,orders,outer_left
19:51:03 [INFO] MapReduceLauncher - detailed locations: M: customers[2,12],customers[-1,-1],outer_left[7,14],orders[4,12],orders[-1,-1],outer_left[7,14] C:  R: outer_left[7,14]
19:51:03 [INFO] MapReduceLauncher - 0% complete
19:51:03 [INFO] MapReduceLauncher - Running jobs are [job_1729236583670_0043]
19:51:10 [INFO] MapReduceLauncher - 50% complete
19:51:10 [INFO] MapReduceLauncher - Running jobs are [job_1729236583670_0043]
19:51:15 [INFO] MapReduceLauncher - Running jobs are [job_1729236583670_0043]
19:51:18 [INFO] MapReduceLauncher - 100% complete
19:51:18 [INFO] SimplePigStats - Script Statistics:

HadoopVersion   PigVersion      UserId  StartedAt       FinishedAt      Features
3.3.6   0.18.0  ubuntu  2026-10-25 19:51:01     2026-10-25 19:51:18     HASH_JOIN

Success!

Job Stats (time in seconds):
JobId   Maps    Reduces MaxMapTime      MinMapTime      AvgMapTime      MedianMapTime   MaxReduceTime   MinReduceTime   AvgReduceTime   MedianReducetime        Alias   Feature Outputs
job_1729236583670_0043  2       1       1       1       1       1       1       1       customers,orders,outer_left     HASH_JOIN       hdfs://localhost:9000/tmp/temp2064518589/tmp1492282389,

Input(s):
Successfully read 7 records from: "/bdt/exp8/customers.txt"
Successfully read 4 records from: "/bdt/exp8/orders.txt"

Output(s):
Successfully stored 8 records (393 bytes) in: "hdfs://localhost:9000/tmp/temp2064518589/tmp1492282389"

Counters:
Total records written : 8
Total bytes written : 393
Spillable Memory Manager spill count : 0
Total bags proactively spilled: 0
Total records proactively spilled: 0

Job DAG:
job_1729236583670_0043


19:51:18 [INFO] MapReduceLauncher - Success!
(1,Ramesh,32,Ahmedabad,2000.0,,,,)
(2,Khilan,25,Delhi,1500.0,101,2009-11-20 00:00:00,2,1560)
(3,kaushik,23,Kota,2000.0,100,2009-10-08 00:00:00,3,1500)
(3,kaushik,23,Kota,2000.0,102,2009-10-08 00:00:00,3,3000)
(4,Chaitali,25,Mumbai,6500.0,103,2008-05-20 00:00:00,4,2060)
(5,Hardik,27,Bhopal,8500.0,,,,)
```

```
Successfully read 7 records from: "/bdt/exp8/customers.txt"
Successfully read 4 records from: "/bdt/exp8/orders.txt"

Output(s):
Successfully stored 4 records (253 bytes) in: "hdfs://localhost:9000/tmp/temp2064518589/tmp-782984090"

Counters:
Total records written : 4
Total bytes written : 253
Spillable Memory Manager spill count : 0
Total bags proactively spilled: 0
Total records proactively spilled: 0

Job DAG:
job_1729236583670_0044


19:51:40 [INFO] MapReduceLauncher - Success!
(2,Khilan,25,Delhi,1500.0,101,2009-11-20 00:00:00,2,1560)
(3,kaushik,23,Kota,2000.0,100,2009-10-08 00:00:00,3,1500)
(3,kaushik,23,Kota,2000.0,102,2009-10-08 00:00:00,3,3000)
(4,Chaitali,25,Mumbai,6500.0,103,2008-05-20 00:00:00,4,2060)
19:51:40 [WARN] SchemaTupleBackend - SchemaTupleBackend has already been initialized
19:51:42 [INFO] MapReduceLauncher - 1 map-reduce job(s) waiting for submission.
19:51:42 [WARN] JobResourceUploader - No job jar file set.  User classes may not be found. See Job or Job#setJar(String).
19:51:43 [INFO] MapReduceLauncher - HadoopJobId: job_1729236583670_0045
19:51:43 [INFO] MapReduceLauncher - Processing aliases customers,orders,outer_full
19:51:43 [INFO] MapReduceLauncher - detailed locations: M: customers[2,12],customers[-1,-1],outer_full[13,14],orders[4,12],orders[-1,-1],outer_full[13,14] C:  R: outer_full[13,14]
19:51:43 [INFO] MapReduceLauncher - 0% complete
19:51:43 [INFO] MapReduceLauncher - Running jobs are [job_1729236583670_0045]
19:51:50 [INFO] MapReduceLauncher - 50% complete
19:51:50 [INFO] MapReduceLauncher - Running jobs are [job_1729236583670_0045]
19:51:55 [INFO] MapReduceLauncher - Running jobs are [job_1729236583670_0045]
19:51:59 [INFO] MapReduceLauncher - 100% complete
19:51:59 [INFO] SimplePigStats - Script Statistics:

HadoopVersion   PigVersion      UserId  StartedAt       FinishedAt      Features
3.3.6   0.18.0  ubuntu  2026-10-25 19:51:40     2026-10-25 19:51:59     HASH_JOIN

Success!

Job Stats (time in seconds):
JobId   Maps    Reduces MaxMapTime      MinMapTime      AvgMapTime      MedianMapTime   MaxReduceTime   MinReduceTime   AvgReduceTime   MedianReducetime        Alias   Feature Outputs
job_1729236583670_0045  2       1       2       2       2       1       1       1       customers,orders,outer_full     HASH_JOIN       hdfs://localhost:9000/tmp/temp2064518589/tmp2096548376,

Input(s):
Successfully read 7 records from: "/bdt/exp8/customers.txt"
Successfully read 4 records from: "/bdt/exp8/orders.txt"

Output(s):
Successfully stored 8 records (393 bytes) in: "hdfs://localhost:9000/tmp/temp2064518589/tmp2096548376"

Counters:
Total records written : 8
Total bytes written : 393
Spillable Memory Manager spill count : 0
Total bags proactively spilled: 0
Total records proactively spilled: 0

Job DAG:
job_1729236583670_0045


19:51:59 [INFO] MapReduceLauncher - Success!
(1,Ramesh,32,Ahmedabad,2000.0,,,,)
(2,Khilan,25,Delhi,1500.0,101,2009-11-20 00:00:00,2,1560)
(3,kaushik,23,Kota,2000.0,100,2009-10-08 00:00:00,3,1500)
(3,kaushik,23,Kota,2000.0,102,2009-10-08 00:00:00,3,3000)
(4,Chaitali,25,Mumbai,6500.0,103,2008-05-20 00:00:00,4,2060)
(5,Hardik,27,Bhopal,8500.0,,,,)
(6,Komal,22,MP,4500.0,,,,)
(7,Muffy,24,Indore,10000.0,,,,)
19:51:59 [INFO] Main - Pig script completed in 59 seconds and 120 milliseconds (59120 ms)

[ screenshot 4 - take it, then press Enter ]
```

```
EXPERIMENT 8 : Pig script to join, group, sort and filter data
Part 5 : FILTER and ORDER BY (sort), LIMIT

$ cat pig/04_filter_sort.pig
-- Experiment 8 | Part 5 : FILTER and ORDER BY (sort)
student_details = LOAD '/bdt/exp8/student_details.txt' USING PigStorage(',')
    AS (id:int, firstname:chararray, lastname:chararray, age:int, phone:chararray, city:chararray);

-- students who belong to Chennai
filter_data = FILTER student_details BY city == 'Chennai';
DUMP filter_data;

-- sort all students by age (descending), then by first name
order_by_data = ORDER student_details BY age DESC, firstname ASC;
DUMP order_by_data;

-- filter + sort + limit together: the 3 youngest students older than 21
older   = FILTER student_details BY age > 21;
by_age  = ORDER older BY age ASC, id ASC;
top3    = LIMIT by_age 3;
DUMP top3;

$ pig -x mapreduce -4 /home/ubuntu/bdt/pig/log4j.properties -f pig/04_filter_sort.pig
2026-10-25 19:52:26,728 INFO pig.ExecTypeProvider: Trying ExecType : LOCAL
2026-10-25 19:52:26,729 INFO pig.ExecTypeProvider: Trying ExecType : MAPREDUCE
2026-10-25 19:52:26,729 INFO pig.ExecTypeProvider: Picked MAPREDUCE as the ExecType
2026-10-25 19:52:26,759 INFO pig.Main: Loaded log4j properties from file: /home/ubuntu/bdt/pig/log4j.properties
19:52:26 [INFO] Main - Apache Pig version 0.18.0 (r1926484) compiled Jun 16 2025, 10:38:21
19:52:26 [INFO] Main - Logging error messages to: /home/ubuntu/bdt/exp8/pig_1792938146757.log
19:52:26 [INFO] HExecutionEngine - Connecting to hadoop file system at: hdfs://localhost:9000
19:52:27 [WARN] PigServer - ATS is disabled since yarn.timeline-service.enabled set to false
19:52:28 [INFO] MapReduceLauncher - 1 map-reduce job(s) waiting for submission.
19:52:28 [WARN] JobResourceUploader - No job jar file set.  User classes may not be found. See Job or Job#setJar(String).
19:52:30 [INFO] MapReduceLauncher - HadoopJobId: job_1792936583670_0046
19:52:30 [INFO] MapReduceLauncher - Processing aliases filter_data,student_details
19:52:30 [INFO] MapReduceLauncher - detailed locations: M: student_details[2,18],student_details[-1,-1],filter_data[6,14] C:  R:
19:52:30 [INFO] MapReduceLauncher - 0% complete
19:52:30 [INFO] MapReduceLauncher - Running jobs are [job_1792936583670_0046]
19:52:37 [INFO] MapReduceLauncher - 50% complete
19:52:37 [INFO] MapReduceLauncher - Running jobs are [job_1792936583670_0046]
19:52:40 [INFO] MapReduceLauncher - 100% complete
19:52:40 [INFO] SimplePigStats - Script Statistics:

HadoopVersion   PigVersion      UserId  StartedAt       FinishedAt      Features
3.3.6   0.18.0  ubuntu  2026-10-25 19:52:27     2026-10-25 19:52:40     FILTER

Success!

Job Stats (time in seconds):
JobId   Maps    Reduces MaxMapTime      MinMapTime      AvgMapTime      MedianMapTime   MaxReduceTime   MinReduceTime   AvgReduceTime   MedianReducetime        Alias   Feature Outputs
job_1792936583670_0046  1       0       1       1       1       0       0       0       filter_data,student_details     MAP_ONLY        hdfs://localhost:9000/tmp/temp-19349538/tmp-2068148324,

Input(s):
Successfully read 8 records (716 bytes) from: "/bdt/exp8/student_details.txt"

Output(s):
Successfully stored 2 records (108 bytes) in: "hdfs://localhost:9000/tmp/temp-19349538/tmp-2068148324"

Counters:
Total records written : 2
Total bytes written : 108
Spillable Memory Manager spill count : 0
Total bags proactively spilled: 0
Total records proactively spilled: 0

Job DAG:
job_1792936583670_0046


19:52:40 [INFO] MapReduceLauncher - Success!
(6,Archana,Mishra,23,9848022335,Chennai)
(8,Bharathi,Nambiayar,24,9848022333,Chennai)
19:52:40 [WARN] SchemaTupleBackend - SchemaTupleBackend has already been initialized
19:52:42 [INFO] MapReduceLauncher - 1 map-reduce job(s) waiting for submission.
19:52:42 [WARN] JobResourceUploader - No job jar file set.  User classes may not be found. See Job or Job#setJar(String).
```

```
HadoopVersion   PigVersion      UserId  StartedAt       FinishedAt      Features
3.3.6   0.18.0  ubuntu  2026-10-25 19:52:40     2026-10-25 19:53:30     ORDER_BY

Success!

Job Stats (time in seconds):
JobId   Maps    Reduces MaxMapTime      MinMapTime      AvgMapTime      MedianMapTime   MaxReduceTime   MinReduceTime   AvgReduceTime   MedianReducetime        Alias   Feature Outputs
job_1792936583670_0047  1       0       1       1       1       0       0       0       student_details MAP_ONLY
job_1792936583670_0048  1       1       1       1       1       1       1       1       order_by_data   SAMPLER
job_1792936583670_0049  1       1       0       0       0       0       1       1       1       1       order_by_data   ORDER_BY        hdfs://localhost:9000/tmp/temp-19349538/tmp-542358473,

Input(s):
Successfully read 8 records (716 bytes) from: "/bdt/exp8/student_details.txt"

Output(s):
Successfully stored 8 records (396 bytes) in: "hdfs://localhost:9000/tmp/temp-19349538/tmp-542358473"

Counters:
Total records written : 8
Total bytes written : 396
Spillable Memory Manager spill count : 0
Total bags proactively spilled: 0
Total records proactively spilled: 0

Job DAG:
job_1792936583670_0047  ->      job_1792936583670_0048,
job_1792936583670_0048  ->      job_1792936583670_0049,
job_1792936583670_0049


19:53:30 [INFO] MapReduceLauncher - Success!
(8,Bharathi,Nambiayar,24,9848022333,Chennai)
(7,Komal,Nayak,24,9848022334,trivendram)
(6,Archana,Mishra,23,9848022335,Chennai)
(5,Trupthi,Mohanthy,23,9848022336,Bhuwaneshwar)
(3,Rajesh,Khanna,22,9848022339,Delhi)
(2,siddarth,Battacharya,22,9848022338,Kolkata)
(4,Preethi,Agarwal,21,9848022330,Pune)
(1,Rajiv,Reddy,21,9848022337,Hyderabad)
19:53:30 [WARN] SchemaTupleBackend - SchemaTupleBackend has already been initialized
19:53:31 [INFO] MapReduceLauncher - 1 map-reduce job(s) waiting for submission.
19:53:31 [WARN] JobResourceUploader - No job jar file set.  User classes may not be found. See Job or Job#setJar(String).
19:53:32 [INFO] MapReduceLauncher - HadoopJobId: job_1792936583670_0050
19:53:32 [INFO] MapReduceLauncher - Processing aliases older,student_details
19:53:32 [INFO] MapReduceLauncher - detailed locations: M: student_details[2,18],student_details[-1,-1],older[14,10] C:  R:
19:53:32 [INFO] MapReduceLauncher - 0% complete
19:53:32 [INFO] MapReduceLauncher - Running jobs are [job_1792936583670_0050]
19:53:39 [INFO] MapReduceLauncher - 12% complete
19:53:39 [INFO] MapReduceLauncher - Running jobs are [job_1792936583670_0050]
19:53:42 [INFO] MapReduceLauncher - 25% complete
19:53:44 [INFO] MapReduceLauncher - 1 map-reduce job(s) waiting for submission.
19:53:44 [WARN] JobResourceUploader - No job jar file set.  User classes may not be found. See Job or Job#setJar(String).
19:53:45 [INFO] MapReduceLauncher - HadoopJobId: job_1792936583670_0051
19:53:45 [INFO] MapReduceLauncher - Processing aliases by_age
19:53:45 [INFO] MapReduceLauncher - detailed locations: M: by_age[15,10] C:  R:
19:53:52 [INFO] MapReduceLauncher - 37% complete
19:53:52 [INFO] MapReduceLauncher - Running jobs are [job_1792936583670_0051]
19:53:57 [INFO] MapReduceLauncher - 50% complete
19:53:57 [INFO] MapReduceLauncher - Running jobs are [job_1792936583670_0051]
19:54:01 [INFO] MapReduceLauncher - 1 map-reduce job(s) waiting for submission.
19:54:01 [WARN] JobResourceUploader - No job jar file set.  User classes may not be found. See Job or Job#setJar(String).
19:54:02 [INFO] MapReduceLauncher - HadoopJobId: job_1792936583670_0052
19:54:02 [INFO] MapReduceLauncher - Processing aliases by_age
19:54:02 [INFO] MapReduceLauncher - detailed locations: M: by_age[15,10] C:  R:
19:54:09 [INFO] MapReduceLauncher - 62% complete
19:54:09 [INFO] MapReduceLauncher - Running jobs are [job_1792936583670_0052]
19:54:14 [INFO] MapReduceLauncher - 75% complete
19:54:14 [INFO] MapReduceLauncher - Running jobs are [job_1792936583670_0052]
19:54:18 [INFO] MapReduceLauncher - 1 map-reduce job(s) waiting for submission.
19:54:18 [WARN] JobResourceUploader - No job jar file set.  User classes may not be found. See Job or Job#setJar(String).
19:54:20 [INFO] MapReduceLauncher - HadoopJobId: job_1792936583670_0053
19:54:20 [INFO] MapReduceLauncher - Processing aliases by_age
19:54:20 [INFO] MapReduceLauncher - detailed locations: M: by_age[15,10] C:  R:
19:54:29 [INFO] MapReduceLauncher - 87% complete
```

```
19:53:32 [INFO] MapReduceLauncher - Processing aliases older,student_details
19:53:32 [INFO] MapReduceLauncher - detailed locations: M: student_details[2,18],student_details[-1,-1],older[14,10] C:  R:
19:53:32 [INFO] MapReduceLauncher - 0% complete
19:53:32 [INFO] MapReduceLauncher - Running jobs are [job_1792936583670_0050]
19:53:39 [INFO] MapReduceLauncher - 12% complete
19:53:39 [INFO] MapReduceLauncher - Running jobs are [job_1792936583670_0050]
19:53:42 [INFO] MapReduceLauncher - 25% complete
19:53:44 [INFO] MapReduceLauncher - 1 map-reduce job(s) waiting for submission.
19:53:44 [WARN] JobResourceUploader - No job jar file set.  User classes may not be found. See Job or Job#setJar(String).
19:53:45 [INFO] MapReduceLauncher - HadoopJobId: job_1792936583670_0051
19:53:45 [INFO] MapReduceLauncher - Processing aliases by_age
19:53:45 [INFO] MapReduceLauncher - detailed locations: M: by_age[15,10] C:  R:
19:53:52 [INFO] MapReduceLauncher - 37% complete
19:53:52 [INFO] MapReduceLauncher - Running jobs are [job_1792936583670_0051]
19:53:57 [INFO] MapReduceLauncher - 50% complete
19:53:57 [INFO] MapReduceLauncher - Running jobs are [job_1792936583670_0051]
19:54:01 [INFO] MapReduceLauncher - 1 map-reduce job(s) waiting for submission.
19:54:01 [WARN] JobResourceUploader - No job jar file set.  User classes may not be found. See Job or Job#setJar(String).
19:54:02 [INFO] MapReduceLauncher - HadoopJobId: job_1792936583670_0052
19:54:02 [INFO] MapReduceLauncher - Processing aliases by_age
19:54:02 [INFO] MapReduceLauncher - detailed locations: M: by_age[15,10] C:  R:
19:54:09 [INFO] MapReduceLauncher - 62% complete
19:54:09 [INFO] MapReduceLauncher - Running jobs are [job_1792936583670_0052]
19:54:14 [INFO] MapReduceLauncher - 75% complete
19:54:14 [INFO] MapReduceLauncher - Running jobs are [job_1792936583670_0052]
19:54:18 [INFO] MapReduceLauncher - 1 map-reduce job(s) waiting for submission.
19:54:18 [WARN] JobResourceUploader - No job jar file set.  User classes may not be found. See Job or Job#setJar(String).
19:54:20 [INFO] MapReduceLauncher - HadoopJobId: job_1792936583670_0053
19:54:20 [INFO] MapReduceLauncher - Processing aliases by_age
19:54:20 [INFO] MapReduceLauncher - detailed locations: M: by_age[15,10] C:  R:
19:54:29 [INFO] MapReduceLauncher - 87% complete
19:54:29 [INFO] MapReduceLauncher - Running jobs are [job_1792936583670_0053]
19:54:32 [INFO] MapReduceLauncher - Running jobs are [job_1792936583670_0053]
19:54:35 [INFO] MapReduceLauncher - 100% complete
19:54:35 [INFO] SimplePigStats - Script Statistics:

HadoopVersion   PigVersion      UserId  StartedAt       FinishedAt      Features
3.3.6   0.18.0  ubuntu  2026-10-25 19:53:30     2026-10-25 19:54:35     ORDER_BY,FILTER,LIMIT

Success!

Job Stats (time in seconds):
JobId   Maps    Reduces MaxMapTime      MinMapTime      AvgMapTime      MedianMapTime   MaxReduceTime   MinReduceTime   AvgReduceTime   MedianReducetime        Alias   Feature Outputs
job_1792936583670_0050  1       0       1       1       1       0       0       0       0       older,student_details   MAP_ONLY
job_1792936583670_0051  1       1       1       1       1       1       1       1       1       by_age  SAMPLER
job_1792936583670_0052  1       1       1       1       1       1       1       1       1       by_age  ORDER_BY,COMBINER
job_1792936583670_0053  1       1       1       1       1       2       2       2       2       by_age          hdfs://localhost:9000/tmp/temp-19349538/tmp802432481,

Input(s):
Successfully read 8 records (716 bytes) from: "/bdt/exp8/student_details.txt"

Output(s):
Successfully stored 3 records (161 bytes) in: "hdfs://localhost:9000/tmp/temp-19349538/tmp802432481"

Counters:
Total records written : 3
Total bytes written : 161
Spillable Memory Manager spill count : 0
Total bags proactively spilled: 0
Total records proactively spilled: 0

Job DAG:
job_1792936583670_0050  ->      job_1792936583670_0051,
job_1792936583670_0051  ->      job_1792936583670_0052,
job_1792936583670_0052  ->      job_1792936583670_0053,
job_1792936583670_0053


19:54:35 [INFO] MapReduceLauncher - Success!
(2,siddarth,Battacharya,22,9848022338,Kolkata)
(3,Rajesh,Khanna,22,9848022339,Delhi)
(5,Trupthi,Mohanthy,23,9848022336,Bhuvaneshwar)
19:54:35 [INFO] Main - Pig script completed in 2 minutes, 9 seconds and 54 milliseconds (129054 ms)

( screenshot 5 - take it, then press Enter )
```

```
EXPERIMENT 8 : Pig script to join, group, sort and filter data
Part 6 : Handwritten questions 1-3: student-department JOIN, salary per department, salary > 50000 sorted

$ cat pig/05_handwritten.pig
-- Experiment 8 | Part 6 : handwritten questions 1-3 executed on sample data
-- Q1: JOIN Student(RollNo, Name, DeptID) with Department(DeptID, DeptName)
students    = LOAD '/bdt/exp8/students.txt'    USING PigStorage(',') AS (rollno:int, name:chararray, deptid:chararray);
departments = LOAD '/bdt/exp8/departments.txt' USING PigStorage(',') AS (deptid:chararray, deptname:chararray);
stud_dept   = JOIN students BY deptid, departments BY deptid;
q1          = FOREACH stud_dept GENERATE students::rollno AS rollno, students::name AS name, departments::deptname AS deptname;
q1_sorted   = ORDER q1 BY rollno;
DUMP q1_sorted;

-- Q2: GROUP employees by department and total the salary of each department
employees = LOAD '/bdt/exp8/employees.txt' USING PigStorage(',') AS (empid:chararray, deptname:chararray, salary:int);
by_dept   = GROUP employees BY deptname;
q2        = FOREACH by_dept GENERATE group AS deptname, SUM(employees.salary) AS total_salary, COUNT(employees) AS employees;
DUMP q2;

-- Q3: FILTER employees with salary > 50000 and SORT by salary descending
high = FILTER employees BY salary > 50000;
q3   = ORDER high BY salary DESC;
DUMP q3;

$ pig -x mapreduce -4 /home/ubuntu/bdt/pig/log4j.properties -f pig/05_handwritten.pig
2026-10-25 19:55:12,008 INFO pig.ExecTypeProvider: Trying ExecType : LOCAL
2026-10-25 19:55:12,009 INFO pig.ExecTypeProvider: Trying ExecType : MAPREDUCE
2026-10-25 19:55:12,009 INFO pig.ExecTypeProvider: Picked MAPREDUCE as the ExecType
2026-10-25 19:55:12,037 INFO pig.Main: Loaded log4j properties from file: /home/ubuntu/bdt/pig/log4j.properties
19:55:12 [INFO] Main - Apache Pig version 0.18.0 (r1926484) compiled Jun 16 2025, 10:38:21
19:55:12 [INFO] Main - Logging error messages to: /home/ubuntu/bdt/exp8/pig_1792938312035.log
19:55:12 [INFO] HExecutionEngine - Connecting to hadoop file system at: hdfs://localhost:9000
19:55:12 [WARN] PigServer - ATS is disabled since yarn.timeline-service.enabled set to false
19:55:14 [INFO] MapReduceLauncher - 1 map-reduce job(s) waiting for submission.
19:55:14 [WARN] JobResourceUploader - No job jar file set.  User classes may not be found. See Job or Job#setJar(String).
19:55:14 [INFO] MapReduceLauncher - HadoopJobId: job_1792936583670_0054
19:55:14 [INFO] MapReduceLauncher - Processing aliases departments,q1,stud_dept,students
19:55:14 [INFO] MapReduceLauncher - detailed locations: M: students[3,14],students[-1,-1],stud_dept[5,14],departments[4,14],departments[-1,-1],stud_dept[5,14] C:  R: q1[6,14]
19:55:14 [INFO] MapReduceLauncher - 0% complete
19:55:14 [INFO] MapReduceLauncher - Running jobs are [job_1792936583670_0054]
19:55:21 [INFO] MapReduceLauncher - 16% complete
19:55:21 [INFO] MapReduceLauncher - Running jobs are [job_1792936583670_0054]
19:55:27 [INFO] MapReduceLauncher - 33% complete
19:55:27 [INFO] MapReduceLauncher - Running jobs are [job_1792936583670_0054]
19:55:31 [INFO] MapReduceLauncher - 1 map-reduce job(s) waiting for submission.
19:55:31 [WARN] JobResourceUploader - No job jar file set.  User classes may not be found. See Job or Job#setJar(String).
19:55:32 [INFO] MapReduceLauncher - HadoopJobId: job_1792936583670_0055
19:55:32 [INFO] MapReduceLauncher - Processing aliases q1_sorted
19:55:32 [INFO] MapReduceLauncher - detailed locations: M: q1_sorted[7,14] C:  R:
19:55:41 [INFO] MapReduceLauncher - 50% complete
19:55:41 [INFO] MapReduceLauncher - Running jobs are [job_1792936583670_0055]
19:55:46 [INFO] MapReduceLauncher - 66% complete
19:55:46 [INFO] MapReduceLauncher - Running jobs are [job_1792936583670_0055]
19:55:49 [INFO] MapReduceLauncher - 1 map-reduce job(s) waiting for submission.
19:55:49 [WARN] JobResourceUploader - No job jar file set.  User classes may not be found. See Job or Job#setJar(String).
19:55:50 [INFO] MapReduceLauncher - HadoopJobId: job_1792936583670_0056
19:55:50 [INFO] MapReduceLauncher - Processing aliases q1_sorted
19:55:50 [INFO] MapReduceLauncher - detailed locations: M: q1_sorted[7,14] C:  R:
19:55:59 [INFO] MapReduceLauncher - 83% complete
19:55:59 [INFO] MapReduceLauncher - Running jobs are [job_1792936583670_0056]
19:56:04 [INFO] MapReduceLauncher - Running jobs are [job_1792936583670_0056]
19:56:05 [INFO] MapReduceLauncher - 100% complete
19:56:05 [INFO] SimplePigStats - Script Statistics:

HadoopVersion   PigVersion      UserId  StartedAt       FinishedAt      Features
3.3.6   0.18.0  ubuntu  2026-10-25 19:55:12     2026-10-25 19:56:05     HASH_JOIN,ORDER_BY

Success!

Job Stats (time in seconds):
JobId   Maps    Reduces MaxMapTime      MinMapTime      AvgMapTime      MedianMapTime   MaxReduceTime   MinReduceTime   AvgReduceTime   MedianReducetime        Alias   Feature Outputs
job_1792936583670_0054  2       1       2       1       2       2       2       2       2       departments,q1,stud_dept,students       HASH_JOIN
job_1792936583670_0055  1       1       1       1       1       1       1       1       1       q1_sorted       SAMPLER
job_1792936583670_0056  1       1       1       1       1       1       1       1       1       q1_sorted       ORDER_BY        hdfs://localhost:9000/tmp/temp1710052857/tmp1785852745,
```

```
3.3.6   0.18.0  ubuntu  2026-10-25 19:55:12     2026-10-25 19:56:05     HASH_JOIN,ORDER_BY

Success!

Job Stats (time in seconds):
JobId   Maps    Reduces MaxMapTime      MinMapTime      AvgMapTime      MedianMapTime   MaxReduceTime   MinReduceTime   AvgReduceTime   MedianReducetime        Alias   Feature Outputs
job_1792936583670_0054  2       1       2       1       2       2       2       2       2       departments,q1,stud_dept,students       HASH_JOIN
job_1792936583670_0055  1       1       1       1       1       1       1       1       1       q1_sorted       SAMPLER
job_1792936583670_0056  1       1       1       1       1       1       1       1       1       q1_sorted       ORDER_BY        hdfs://localhost:9000/tmp/temp1710052857/tmp1785852745,

Input(s):
Successfully read 6 records from: "/bdt/exp8/students.txt"
Successfully read 3 records from: "/bdt/exp8/departments.txt"

Output(s):
Successfully stored 5 records (174 bytes) in: "hdfs://localhost:9000/tmp/temp1710052857/tmp1785852745"

Counters:
Total records written : 5
Total bytes written : 174
Spillable Memory Manager spill count : 0
Total bags proactively spilled: 0
Total records proactively spilled: 0

Job DAG:
job_1792936583670_0054  ->      job_1792936583670_0055,
job_1792936583670_0055  ->      job_1792936583670_0056,
job_1792936583670_0056


19:56:05 [INFO] MapReduceLauncher - Success!
(1,Aarav,Computer Engineering)
(2,Diya,Information Technology)
(3,Kabir,Computer Engineering)
(4,Meera,Data Science)
(5,Rohan,Information Technology)
19:56:07 [INFO] MapReduceLauncher - 1 map-reduce job(s) waiting for submission.
19:56:07 [WARN] JobResourceUploader - No job jar file set.  User classes may not be found. See Job or Job#setJar(String).
19:56:08 [INFO] MapReduceLauncher - HadoopJobId: job_1792936583670_0057
19:56:08 [INFO] MapReduceLauncher - Processing aliases by_dept,employees,q2
19:56:08 [INFO] MapReduceLauncher - detailed locations: M: employees[11,12],employees[-1,-1],q2[13,12],by_dept[12,12] C: q2[13,12],by_dept[12,12] R: q2[13,12]
19:56:08 [INFO] MapReduceLauncher - 0% complete
19:56:08 [INFO] MapReduceLauncher - Running jobs are [job_1792936583670_0057]
19:56:17 [INFO] MapReduceLauncher - 50% complete
19:56:17 [INFO] MapReduceLauncher - Running jobs are [job_1792936583670_0057]
19:56:20 [INFO] MapReduceLauncher - Running jobs are [job_1792936583670_0057]
19:56:23 [INFO] MapReduceLauncher - 100% complete
19:56:23 [INFO] SimplePigStats - Script Statistics:

HadoopVersion   PigVersion      UserId  StartedAt       FinishedAt      Features
3.3.6   0.18.0  ubuntu  2026-10-25 19:56:05     2026-10-25 19:56:23     GROUP_BY

Success!

Job Stats (time in seconds):
JobId   Maps    Reduces MaxMapTime      MinMapTime      AvgMapTime      MedianMapTime   MaxReduceTime   MinReduceTime   AvgReduceTime   MedianReducetime        Alias   Feature Outputs
job_1792936583670_0057  1       1       1       1       1       1       1       1       1       by_dept,employees,q2    GROUP_BY,COMBINER       hdfs://localhost:9000/tmp/temp1710052857/tmp-159795735,

Input(s):
Successfully read 8 records (488 bytes) from: "/bdt/exp8/employees.txt"

Output(s):
Successfully stored 4 records (69 bytes) in: "hdfs://localhost:9000/tmp/temp1710052857/tmp-159795735"

Counters:
Total records written : 4
Total bytes written : 69
Spillable Memory Manager spill count : 0
Total bags proactively spilled: 0
Total records proactively spilled: 0

Job DAG:
job_1792936583670_0057
```

```
19:56:23 [INFO] MapReduceLauncher - Success!
(HR,100000,2)
(IT,193000,3)
(Sales,39000,1)
(Finance,130000,2)
19:56:24 [INFO] MapReduceLauncher - 1 map-reduce job(s) waiting for submission.
19:56:24 [WARN] JobResourceUploader - No job jar file set.  User classes may not be found. See Job or Job#setJar(String).
19:56:25 [INFO] MapReduceLauncher - HadoopJobId: job_1792936583670_0058
19:56:25 [INFO] MapReduceLauncher - Processing aliases employees,high
19:56:25 [INFO] MapReduceLauncher - detailed locations: M: employees[11,12],employees[-1,-1],high[17,7] C:  R:
19:56:25 [INFO] MapReduceLauncher - 0% complete
19:56:25 [INFO] MapReduceLauncher - Running jobs are [job_1792936583670_0058]
19:56:35 [INFO] MapReduceLauncher - 16% complete
19:56:35 [INFO] MapReduceLauncher - Running jobs are [job_1792936583670_0058]
19:56:40 [INFO] MapReduceLauncher - 33% complete
19:56:42 [INFO] MapReduceLauncher - 1 map-reduce job(s) waiting for submission.
19:56:42 [WARN] JobResourceUploader - No job jar file set.  User classes may not be found. See Job or Job#setJar(String).
19:56:43 [INFO] MapReduceLauncher - HadoopJobId: job_1792936583670_0059
19:56:43 [INFO] MapReduceLauncher - Processing aliases q3
19:56:43 [INFO] MapReduceLauncher - detailed locations: M: q3[18,7] C:  R:
19:56:50 [INFO] MapReduceLauncher - 50% complete
19:56:50 [INFO] MapReduceLauncher - Running jobs are [job_1792936583670_0059]
19:56:55 [INFO] MapReduceLauncher - 66% complete
19:56:55 [INFO] MapReduceLauncher - Running jobs are [job_1792936583670_0059]
19:57:00 [INFO] MapReduceLauncher - 1 map-reduce job(s) waiting for submission.
19:57:00 [WARN] JobResourceUploader - No job jar file set.  User classes may not be found. See Job or Job#setJar(String).
19:57:00 [INFO] MapReduceLauncher - HadoopJobId: job_1792936583670_0060
19:57:00 [INFO] MapReduceLauncher - Processing aliases q3
19:57:00 [INFO] MapReduceLauncher - detailed locations: M: q3[18,7] C:  R:
19:57:10 [INFO] MapReduceLauncher - 83% complete
19:57:10 [INFO] MapReduceLauncher - Running jobs are [job_1792936583670_0060]
19:57:13 [INFO] MapReduceLauncher - Running jobs are [job_1792936583670_0060]
19:57:16 [INFO] MapReduceLauncher - 100% complete
19:57:16 [INFO] SimplePigStats - Script Statistics:

HadoopVersion   PigVersion      UserId  StartedAt       FinishedAt      Features
3.3.6   0.18.0  ubuntu  2026-10-25 19:56:23     2026-10-25 19:57:16     ORDER_BY,FILTER

Success!

Job Stats (time in seconds):
JobId   Maps    Reduces MaxMapTime      MinMapTime      AvgMapTime      MedianMapTime   MaxReduceTime   MinReduceTime   AvgReduceTime   MedianReducetime        Alias   Feature Outputs
job_1792936583670_0058  1       0       1       1       1       1       0       0       0       employees,high  MAP_ONLY
job_1792936583670_0059  1       1       1       1       1       1       1       1       1       q3      SAMPLER
job_1792936583670_0060  1       1       1       1       1       1       1       1       1       q3      ORDER_BY        hdfs://localhost:9000/tmp/temp1710052857/tmp-968104090,

Input(s):
Successfully read 8 records (488 bytes) from: "/bdt/exp8/employees.txt"

Output(s):
Successfully stored 5 records (105 bytes) in: "hdfs://localhost:9000/tmp/temp1710052857/tmp-968104090"

Counters:
Total records written : 5
Total bytes written : 105
Spillable Memory Manager spill count : 0
Total bags proactively spilled: 0
Total records proactively spilled: 0

Job DAG:
job_1792936583670_0058  ->      job_1792936583670_0059,
job_1792936583670_0059  ->      job_1792936583670_0060,
job_1792936583670_0060


19:57:16 [INFO] MapReduceLauncher - Success!
(E04,IT,81000)
(E03,Finance,72000)
(E01,IT,65000)
(E07,Finance,58000)
(E06,HR,52000)
19:57:16 [INFO] Main - Pig script completed in 2 minutes, 4 seconds and 682 milliseconds (124682 ms)

[ screenshot 6 - take it, then press Enter ]
```

Handwritten Questions

1. Consider two datasets, Student(RollNo, Name, DeptID) and Department(DeptID, DeptName). Write a Pig Latin script to perform a JOIN operation and display each student's name along with the

department name.

2. Consider an employee dataset containing EmpID, DeptName, and Salary. Write a Pig Latin script to GROUP employees by department and calculate the total salary for each department.

3. Write a Pig Latin script to FILTER employees whose salary is greater than ₹50,000 and SORT the resulting records in descending order of salary.

4. Explain the difference between JOIN and GROUP operations in Apache Pig. Illustrate your answer with a suitable dataset and the expected output of each operation.

@1)

Join studend and department database

```
student = Load 'student.txt' using PigStorage
          (',') As (Roll no: Int, Name: Chararray,
                    deptID : Int);

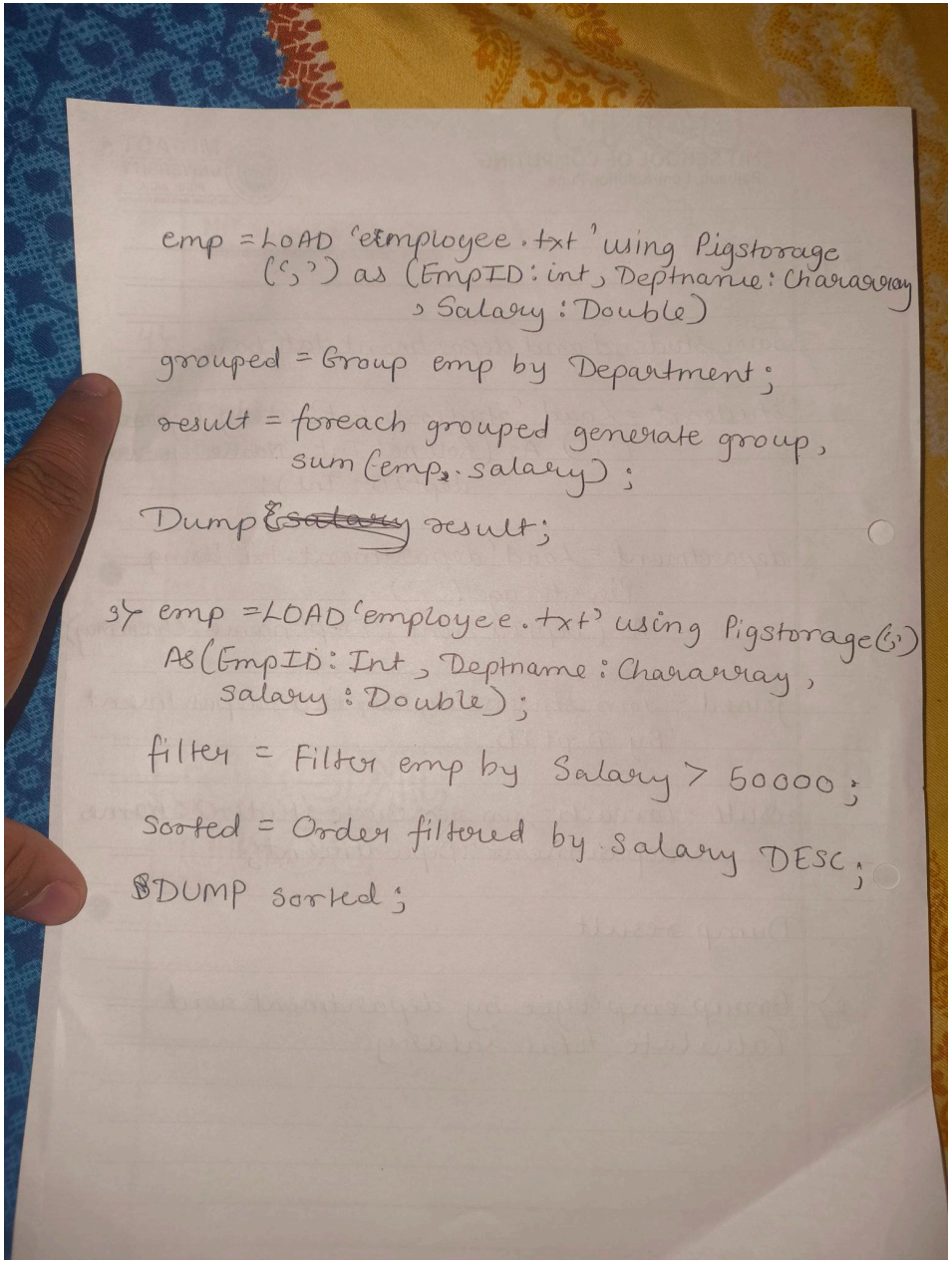
department = Load 'department.txt' Using
             Pig storage (',')
             As (DeptId : uint, Deptname : Chararray)

joined = Join student By DeptID, department
         By DeptID;

result = Foreach join generate student : Name,
         department ::Department;

Dump result
```

2) Group employee by department and Calculate total salary.



```
emp = LOAD 'employee.txt' using PigStorage
      (',') as (EmpID: int, Deptname: Chararray
      , Salary : Double)

grouped = Group emp by Department;

result = foreach grouped generate group,
         sum(emp.salary);

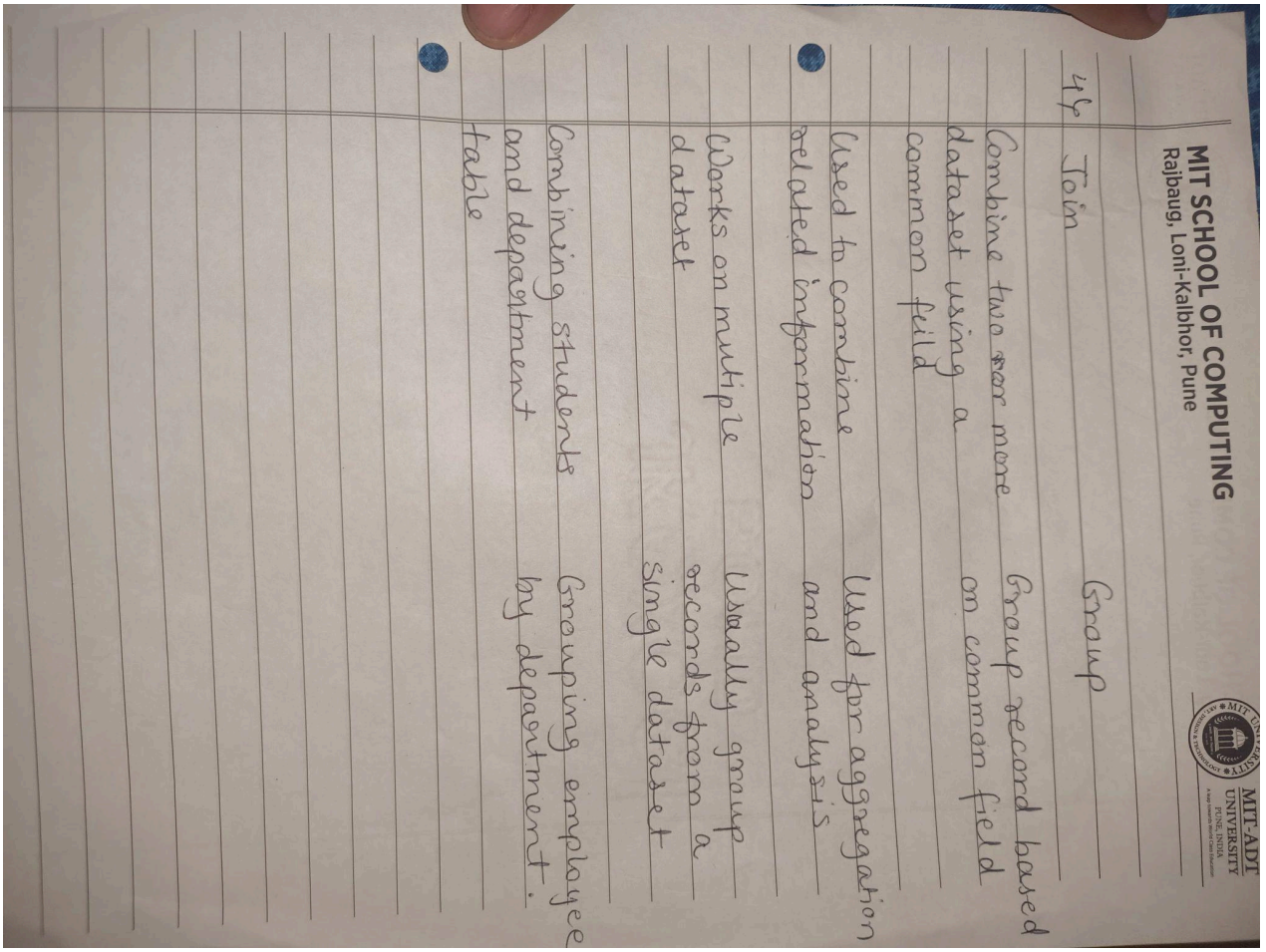
Dump result;

// emp = LOAD 'employee.txt' using PigStorage(',')
   As(EmpID: Int, Deptname : Chararray,
      Salary : Double);

filter = Filter emp by Salary > 50000;

Sorted = Order filtered by Salary DESC;

$DUMP Sorted;
```



Acceptance Criteria

1 The student correctly loads structured datasets and defines appropriate schemas for the input relations. 2 The student implements the JOIN operation correctly and produces the expected combined records.

3 The student successfully performs GROUP and FILTER operations and obtains the expected grouped and filtered results.

4 The student executes the SORT operation correctly, verifies the output order, and explains the results of the relational operations.

Conclusion: Pig Latin scripts were written to join, group, sort and filter data stored in HDFS. GROUP with COUNT and SUM, inner and outer joins, FILTER and ORDER BY all ran successfully as MapReduce jobs and gave the expected results. Thus, Pig provides simple SQL-like operators for processing large datasets on Hadoop.